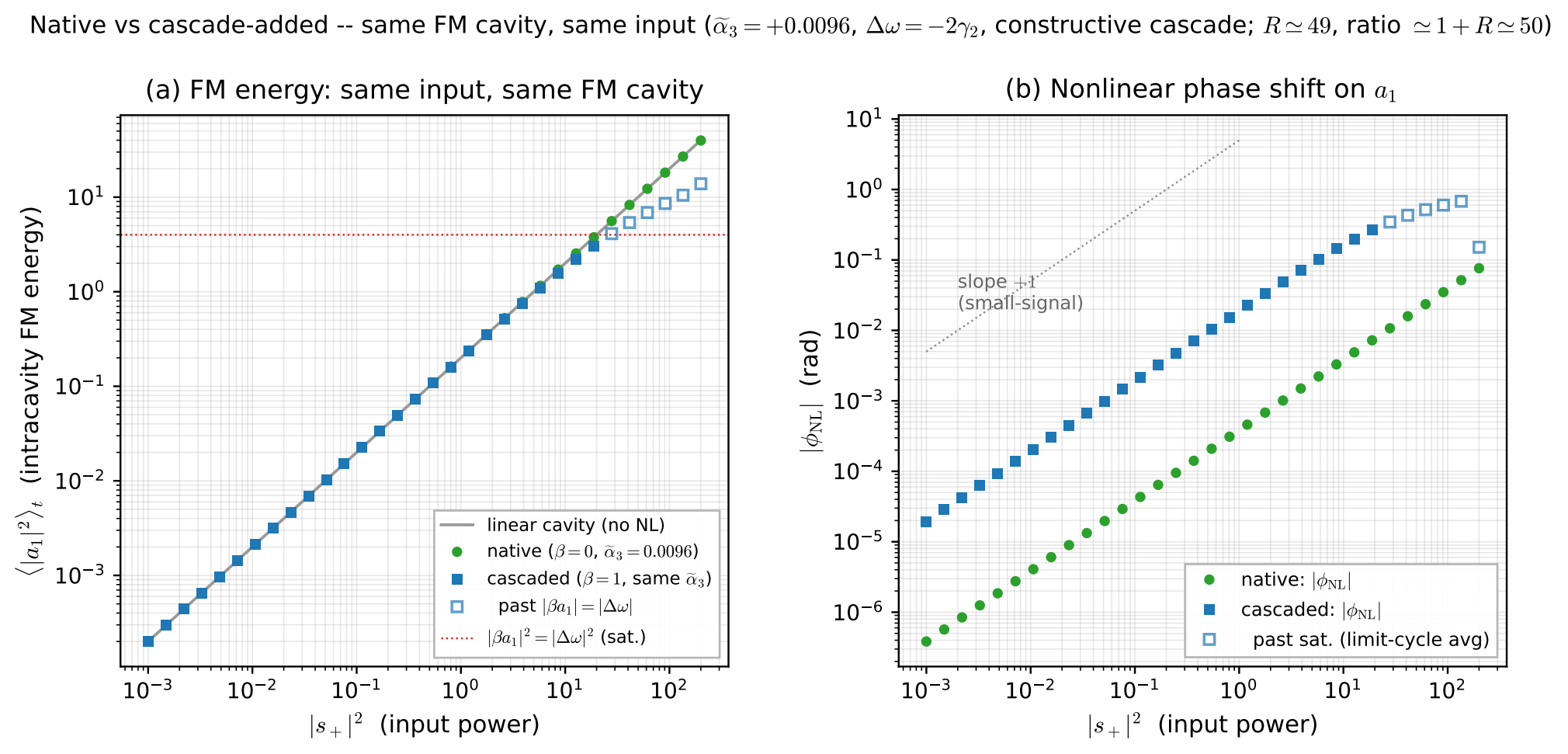
Native vs cascade-added -- same FM cavity, same input ($\widetilde{\alpha}_3 = +0.0096$, $\Delta\omega = -2\gamma_2$, constructive cascade; $R \simeq 49$, ratio $\simeq 1 + R \simeq 50$)
(a) FM energy: same input, same FM cavity
$\langle |a_1|^2 \rangle_t$ (intracavity FM energy)
$|s_+|^2$ (input power)
linear cavity (no NL)
native ($\beta = 0$, $\widetilde{\alpha}_3 = 0.0096$)
cascaded ($\beta = 1$, same $\widetilde{\alpha}_3$)
past $|\beta a_1| = |\Delta\omega|$
$|\beta a_1|^2 = |\Delta\omega|^2$ (sat.)
(b) Nonlinear phase shift on a_1
$|\phi_{\mathrm{NL}}|$ (rad)
slope +1
(small-signal)
native: $|\phi_{\mathrm{NL}}|$
cascaded: $|\phi_{\mathrm{NL}}|$
past sat. (limit-cycle avg)
$|s_+|^2$ (input power)